Supplementary Appendix for Social Impacts of New Radio Markets in Ghana: A Dynamic Structural Analysis

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A.1 Additional Figures and Tables

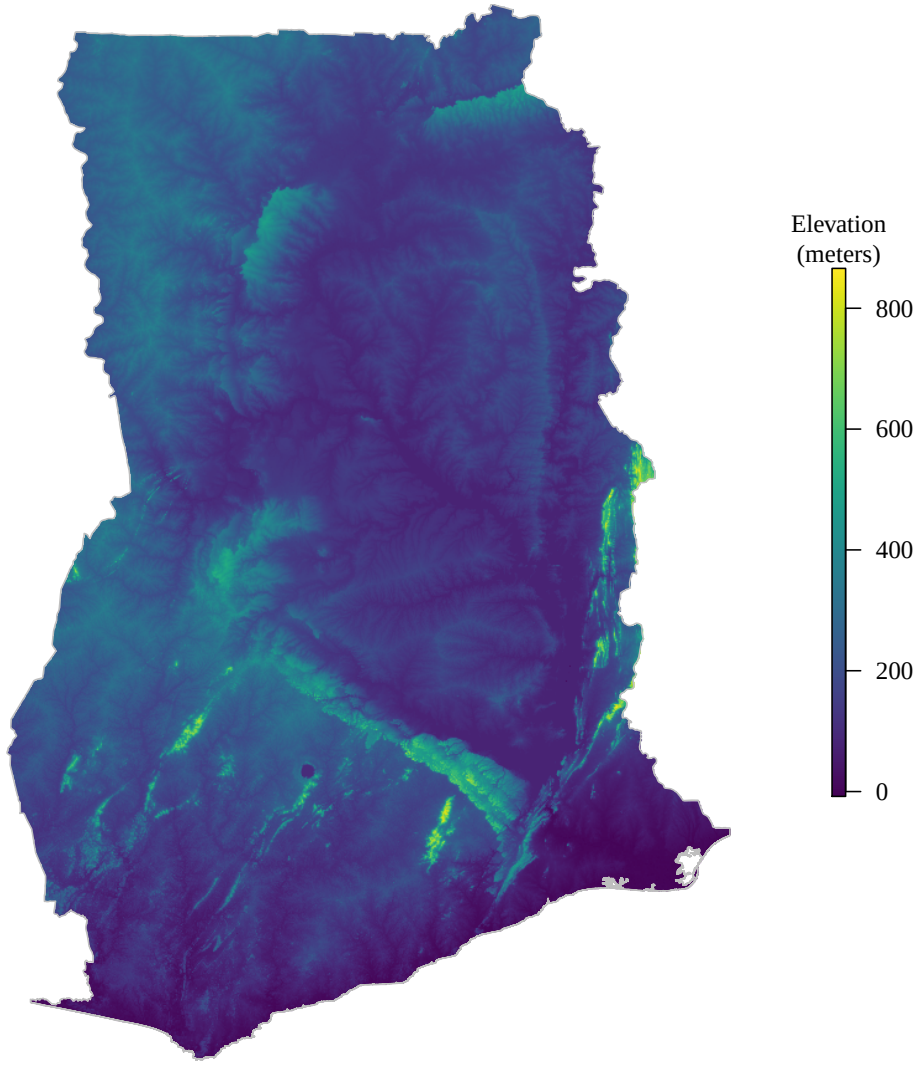
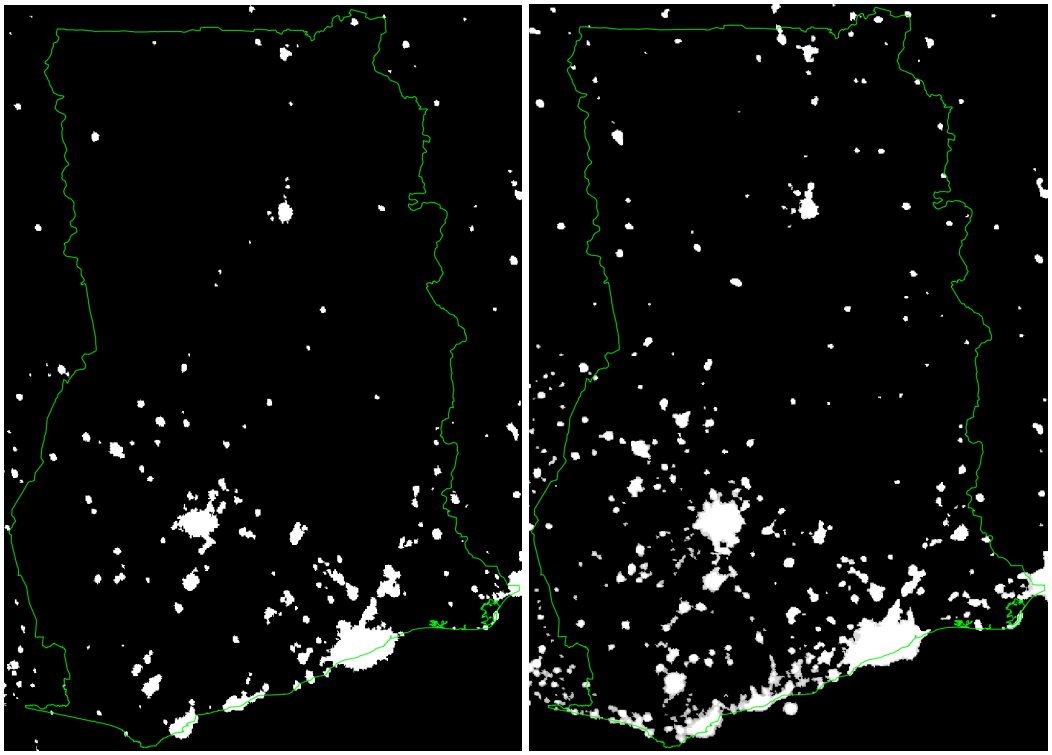


FIGURE A.1: Elevation data from the Shuttle Radar Topography Mission.



(A) 1993 (B) 2013

FIGURE A.2: Nighttime luminosity in Ghana in 1993 and 2013.

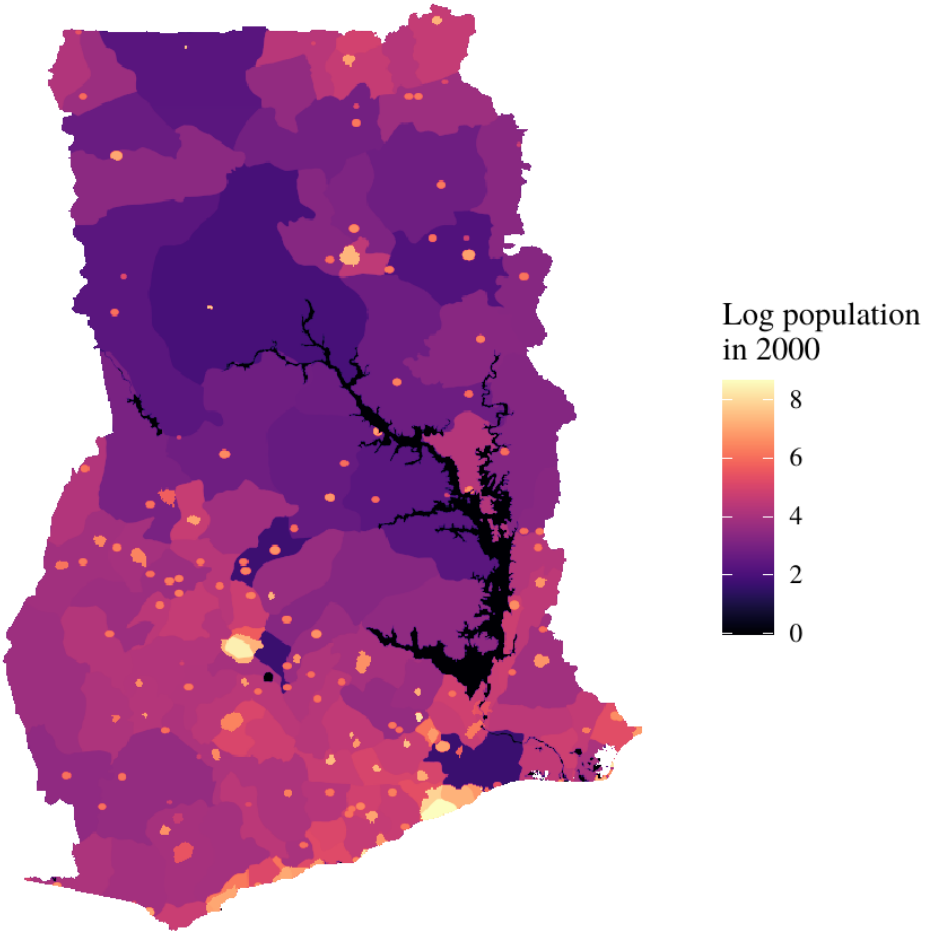
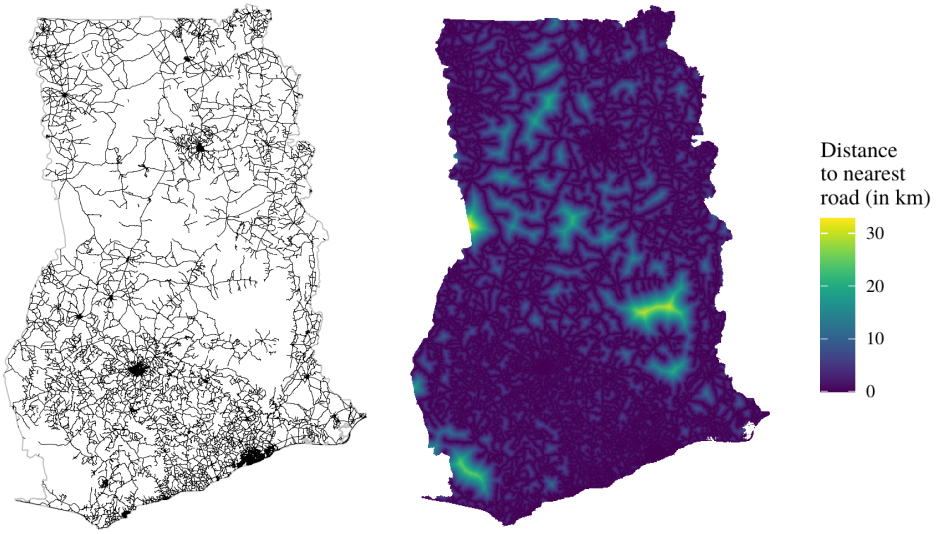


FIGURE A.3: Log population in 2000 from NASA SEDAC.



(A) Road network. (B) Distance to nearest road (in km).

FIGURE A.4: Road network data from OpenStreetMap.

Figure A.4a plots the road network data from OpenStreetMap. For each cell in the coverage data, I calculate the distance in kilometers to the nearest road. These values are shown in Figure A.4b.

	Dependent variable:		
	Initial nighttime luminosity	Elevation range	Distance to nearest road
	(1)	(2)	(3)
Coverage	0.007	−0.576	−0.190
	(0.005)	(0.412)	(0.176)
Mean dependent variable	0.015	27.568	3.054
Close to random coverage only	Yes	Yes	Yes
Observations	313,326	313,641	313,641

Standard errors are clustered at the district level (137 districts). The dependent variable in column (1) is the value of nighttime luminosity in 1992. The dependent variable in column (2) is the difference between the highest and lowest elevation value (at a 90m×90m resolution) in a 9km×9km grid surrounding the observation. The dependent variable in column (3) is the distance to nearest road (in km) using OpenStreetMap data. The independent variable, *Coverage*, is an indicator for whether or not the cell has coverage from any radio station in that year. Observations included are within 1km of the border of a coverage streak.

TABLE A.1: Robustness Checks: Balance Tests.

	Dependent variable:	
	Change in malaria incidence	Change in nighttime luminosity
	(1)	(2)
Future coverage	−0.005	−0.004
	(0.012)	(0.008)
Mean dependent variable	−0.018	0.001
Close to random coverage only	Yes	Yes
Observations	53,374	110,375

Standard errors are clustered at the district level (137 districts). The dependent variable in column (1) is the change in the malaria parasite rate among 2-10 year olds over five years. The dependent variable in column (2) is the change in nighttime luminosity over five years. The independent variable (future coverage) is an indicator for whether or not the cell has coverage ten years in the future. Observations included are within 1km of the border of a coverage streak and did not have coverage before or during the timespan of the dependent variable.

TABLE A.2: Robustness Checks: Placebo Tests.

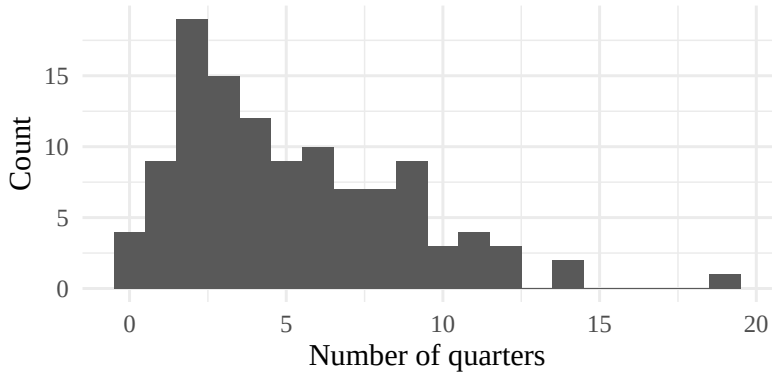
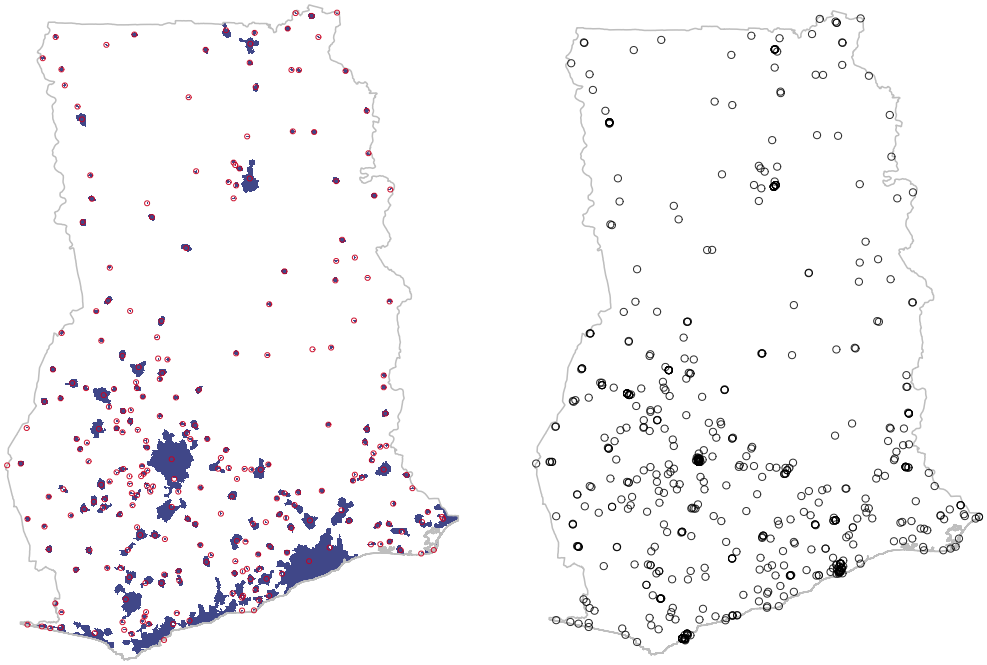


FIGURE A.5: Number of quarters between first authorization and date first active.



(A) Cluster centroids. (B) All potential entrants.

FIGURE A.6: Potential entrant locations.

Figure A.6a plots the areas with positive nighttime luminosity values in 2013 in blue and the red circles are the centroids of nighttime luminosity clusters. These are the additional potential entrants used in estimation. The full set of potential entrant locations is shown in Figure A.6b, which combines the actual station locations and the cluster centroids.

A.2 Additional Information on Radio Stations

A.2.1 License Type Definitions

The National Communications Authority describes each of these types as follows. *Commercial* stations are those that are privately owned, controlled and operated for profit by independent commercial groups or individuals. *Public* stations are stations that are owned and operated by the Ghana Broadcasting Corporation (GBC) or any other stations established by the Government of Ghana whereas *Public Foreign* stations are stations established by foreign governments, such as the BBC. *Community* stations are non-profit and provide service for a specific marginalized community. Ownership and management of community stations are representative of the community. Finally, *Campus* stations are stations operating within the ambit of educational institutions.

A.2.2 Station Language and Content

I searched for information on the stations in our data set through a number of sources, such as their websites, Facebook pages, and live streams. I was able to ascertain the primary languages for 197 of our 436 stations. Table A.3 shows the frequency of different languages among these stations. 107 of these had English as the only language and a further 66 stations were multilingual with Enlish as one of the languages. Akan, Twi and Ewe are the most common languages other than English. There were 13 stations with other languages that no other station broadcasted in. The total sum of stations in Table A.3 adds up to more than 197 because of multilingual stations.

Language	No. of stations	Language	No. of stations
English only	107	Wale	4
Partially English	66	Sissali	4
Akan	22	Gonja	4
Twi	21	Briffor	3
Ewe	20	Kokomba	2
Dagare	9	Hausa	2
Fante	8	Efutu	2
Dagbani	5	Other languages	13

TABLE A.3: Station languages.

I was also able to obtain a short description of the station's genre for 153 stations. 103 of these stations describe themselves as partly music or entertainment stations whereas 91 describe themselves as news or talk stations. 64 of the stations describe themselves as doing both news and music. Therefore music and talk stations make up 85% of all stations I have information on. There are 12 religious stations out of these 153 stations, which is the most common alternative type of station. A smaller number of stations detailed the music genres the played, with most broadcasting several genres. The most common genres are pop, local and African music.

A.3 Example Coverage Streak

Figure A.7 shows an example of a part of Ghana that was included in the reduced-form regressions. The top-left figure shows the cells with radio coverage in that year. Each square in the figure is a cell and each cell is approximately 900m$\times$900m in area. The top-center figure shows the cells that form part of a coverage streak. These are cells where the coverage-land ratio is below 0.2 and the cells are in the

outer quintile of distance away from a station's transmitter. Locations that are included in the regression are shown in the top-right figure. This includes the coverage streak itself (in purple), as well as tiles that do not have coverage but are are within 1km of the edge of the coverage streak (in yellow).

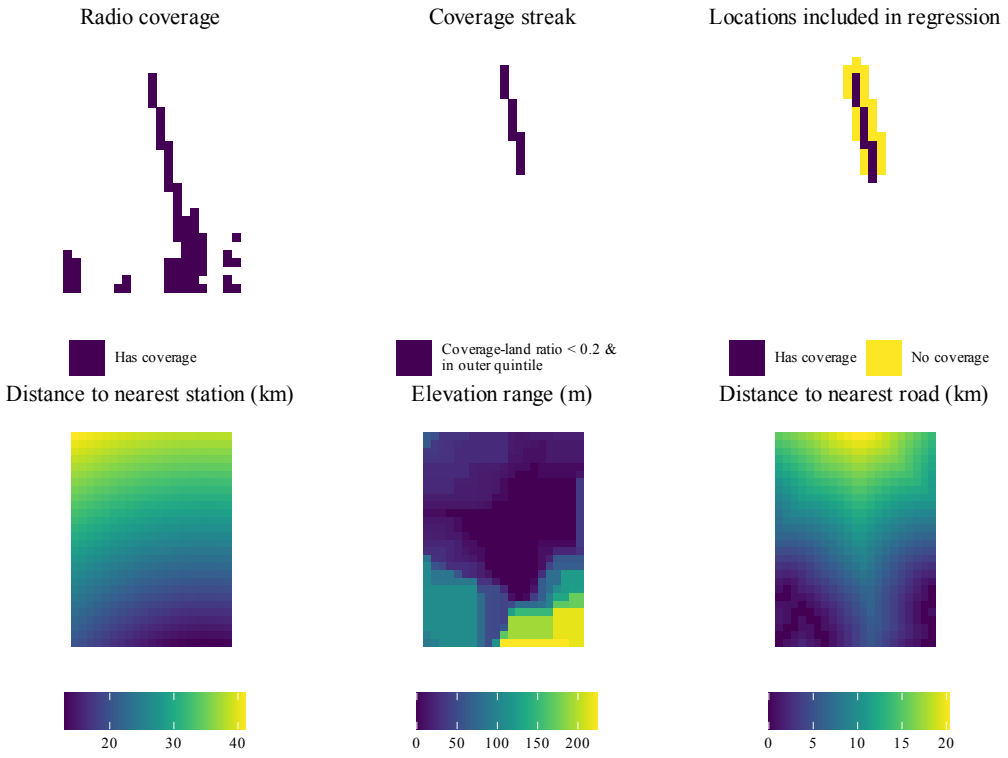


FIGURE A.7: Example coverage streak

If the distance to the nearest station is less than 10km or the maximum elevation range within a 9km×9km grid around the cell is larger than 50, then the cell is omitted from the regression. This is because locations that are close to the source of coverage are more likely to be within the intended coverage area of the station. We also omit locations that are very close to abrupt elevation changes as this may affect outcomes directly. The distance to the nearest station and the elevation range within

a 9km×9km grid in the same area are shown in the bottom-left and bottom-center panels. From this we can see that the coverage streak satisfies both conditions. For comparison purposes, the distance to the nearest road in kilometers is also shown in the bottom-right panel. We can see that the coverage streak is approximately 10km from the nearest road.

A.4 Demographic Health Surveys

Mosquito bed nets are a simple but highly effective method to prevent malaria. To investigate if individuals in locations receiving radio coverage are also more likely to use mosquite bed nets, I make use of the Demographic Health Survey (DHS) data. The DHS program has conducted more than 300 surveys of population, health and nutrition in over 90 countries. In Ghana, they have conducted six repeated cross sections from 1988 to 2014. However, the questions vary year-by-year and information on mosquito bed net usage is contained only in the surveys since 2003. The DHS provides approximate coordinates of each survey cluster, which allows matching with the coverage data. The 1,250 cluster locations are shown in Figure A.8. We can see that the survey locations are spread throughout the entire country. To preserve the anonymity of the survey respondents, the DHS purposefully adds an error of up to 2km for urban clusters and 5km for rural clusters. Thus matching the coverage data with the DHS data using the coordinates that they provide introduces large measurement error. To reduce this measurement error I take the average amount of coverage available within a radius around the cluster locations, rather than taking the number of available radio stations at the coordinates given in the data. The radius I use for each survey cluster corresponds to the maximum possible

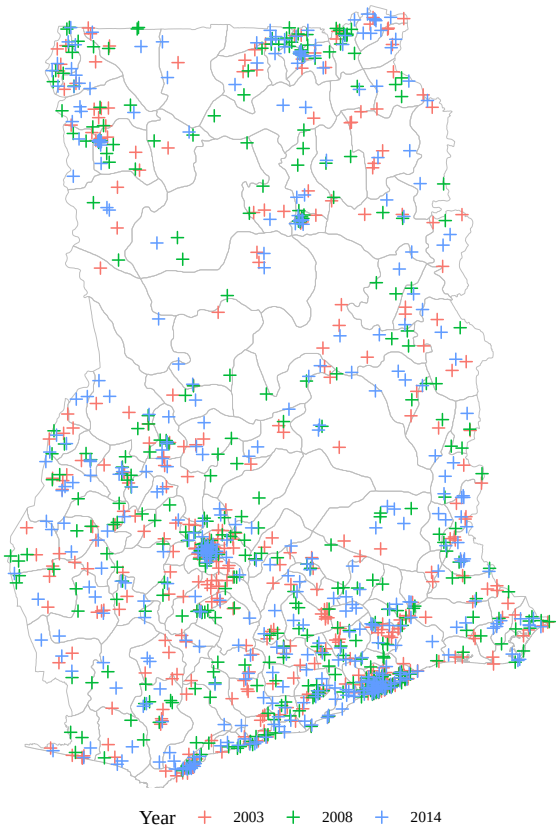


FIGURE A.8: Demographic Health Survey cluster locations by year.

error introduced by the DHS (2km for urban clusters and 5km for rural clusters).

Given the anonymization of survey locations, I cannot implement the coverage streak identification strategy because we do not know if a survey cluster is inside or outside of a coverage streak. Even without this anonymization, there are very few survey clusters that are near coverage streaks. Therefore I use both a fixed effects and an instrumental variables estimation strategy. One possible instrument would be to use local hilliness as this is negatively correlated with radio coverage. However, using such an instrument is problematic if hilliness affects outcomes directly.

For example, areas that are hilly could be more disconnected from society and as a result obtain less information from other villages. Instead of using local hilliness as an instrument, I use distant hilliness. Elevation changes far away from the village can also affect the amount of coverage the village receives. This is because it does not matter if the obstruction between the radio tower and the village is closer to the village or further away and closer to the radio tower: both will affect the signal reaching a village in the same way. However, the distant elevation from the village is less likely to affect the village directly compared to local elevation. Based on this idea, I construct the instrument as follows. I draw two circles around each survey cluster in the DHS data with radii 10km and 15km respectively. These two circles form a ring. I then take the standard deviation of elevation values within the ring as a measure of distant hilliness. Figure A.9 shows an example ring around a survey cluster on top of an elevation heatmap.

Table A.4 shows the regression results. Because coverage only varies at the cluster-year level, I first aggregate the household data to that level. The first column shows the results from an linear model regressing the proportion of households where children use a mosquito bed net on radio station coverage. I control for year fixed effects, region fixed effects and a host of demographic controls. I also control for local malaria presence using the average incidence within a 5km radius of the survey cluster. Coverage increases the probability of mosquito bed net usage on average by 4.2 percentage points. The second column shows the results from the first stage of the instrumental variables regression. The standard deviation of elevation in the ring around the survey cluster has a significant negative effect on coverage. In the instrumental variables regression, coverage on average increases

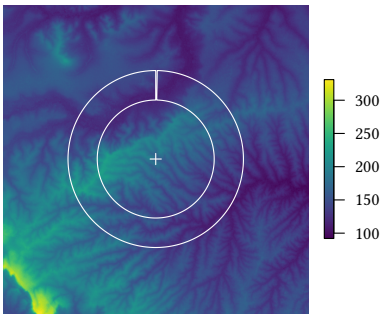


FIGURE A.9: Elevation Instrument

Example ring around a survey cluster overlaid on an elevation heatmap (measured in meters). The instrument for coverage is the standard deviation of elevation within the ring, measuring distant hilliness from a survey cluster.

	Children sleep under bed net OLS (1)	Coverage from commercial station OLS (2)	Children sleep under bed net IV (3)
Coverage from commercial station	0.042		0.169
	(0.020)		(0.072)
Std. dev. of elevation in ring around cluster		−0.003	
		(0.000)	
Demographic controls	Yes	Yes	Yes
Region Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N	1142	1142	1142
Adjusted R^2	0.389	0.504	0.367
F-statistic	25.250	39.607	

Robust standard errors in parentheses. Data is collapsed at the survey cluster-year level. Demographic controls include gender, age, education attainment, religion, ethnicity, literacy, newspaper readership, television viewership, urban/rural. Local malaria presence is also controlled for using Malaria Atlas Project data.

TABLE A.4: DHS Regression Results.

the probability of mosquito bed net usage by 16.9 percentage points. This result is consistent with the results found for malaria prevalence and provides a supporting mechanism. Areas receiving radio coverage are informed of ways to reduce the risk of malaria, and more households use mosquito bed nets. This in turn leads to a lower prevalence rate in those areas.

A.5 Approximating the Number of Listeners and Expected Value Function

A.5.1 Listener Share

Calculating station i's number of listeners at time t, given by

$$M_{it} = \sum_{x \in X} \sum_{y \in Y} m_{it}(x,y) = \sum_{x \in X} \sum_{y \in Y} \frac{f_{it}(x,y)\, p_t(x,y)}{1 + \sum_{j \neq i}^{N} f_{jt}(x,y) + b_t(x,y)}$$

is computationally intensive as it requires calculating $m_{it}(x,y)$ over a large number of locations (x,y). As this needs to be computed many times, I use an approximation $\widetilde{M}_{it}$. This approximation is made up of a number of elements which can be calculated once ahead of time and avoids having to recalculate each $m_{it}(x,y)$ individually. Define the $N \times N$ matrix W with elements:

$$W_{ij} = \sum_{x \in X} \sum_{y \in Y} \frac{\widetilde{f}_i(x,y)\, \widetilde{f}_j(x,y)}{\widetilde{f}_i(x,y)}$$

where $\widetilde{f}_i(x,y)$ is an indicator if station i's coverage would be available at coordinates (x,y) if it were active, and similarly for station j. This is the *overlap matrix*, where element W_{ij} is the share of station i's coverage area that would also be occupied by station j, if both were active. This matrix only needs to be calculated once before estimation, as it does not depend on the entry status of the stations.

Define the monopolist listenership as:

$$P_{it} = \sum_{x \in X} \sum_{y \in Y} \widetilde{f}_i(x,y)\, p_t(x,y)$$

As $p_t(x,y)$ is assumed to grow exogenously and $\widetilde{f}_i(x,y)$ is fixed, P_{it} can be calculated once for each (i,t) and then scaled for changes in population. Further, define the (potential) sum of shares of overlap with public radio stations for station i as:

$$B_{it} = \frac{\sum_{x\in X}\sum_{y\in Y}\widetilde{f}_{it}(x,y)\, b_t(x,y)}{\sum_{x\in X}\sum_{y\in Y}\widetilde{f}_{it}(x,y)}$$

This is a measure of the level of competition from public radio stations for station i at time t which is calculated ahead of time. Finally, define $\hat{a}_t = (\hat{a}_{1t},\ldots,\hat{a}_{Nt})$ as a vector of indicators for each of the N stations being active at time t. With these objects, M_{it} can be approximated with:

$$\widetilde{M}_{it} = \frac{\hat{a}_{it}P_{it}}{W_i\hat{a}_t + B_{it}}$$

where W_i is the ith row of W. For values of M_{it} and $\widetilde{M}_{it}$ observed in the data, the correlation is over 0.96 and the means and standard deviations differ by only 5% and 3% respectively. For estimation, $\widetilde{M}_{it}$ is used in place of M_{it}.

A.5.2 Expected Value Function

To obtain the value of station i entering in period $T-1$ for a given vector of parameters θ and an initial guess of the rival stations' strategies, we need to calculate:

$$\widetilde{v}^1_{iT-1}(s_{T-1},\theta) = \gamma - \theta_{EC} + \lambda\mathbb{E}\left[\beta\,\frac{\pi_{iT}(s_T,\theta)}{1-\beta}\middle| s_{T-1}\right] \tag{1}$$

The expectation in equation (1) is over all possible states s_T. Given the large number of firms, the complete state space is too large to compute this exactly. Rather than

calculating this expectation over every possible state, I calculate the expectation by drawing a sample of 10,000 states for each firm. I have tested this approach with a smaller state space and have found it to produce constitent approximations.

A.6 Profit Function Specification

The model's profit function specification is:

$$\pi_{it}(s_t, \theta) = \theta_M M_{it} + \theta_T t$$

This specification has the feature that a station broadcasting to a location with zero listeners would eventually make positive profits sufficiently far in the future. A possible alternative profit function specification would instead be:

$$\pi_{it}(s_t, \theta) = \theta_M M_{it} e^{\theta_T t}$$

which would not have this feature. Another example would be

$$\pi_{it}(s_t, \theta) = \theta_M M_{it}(1 + \theta_T t)$$

I have experimented with both of these alternative specifications, but in each case the model's fit was worse than the current specification. There are also no instances in the data nor in the counterfactual simulations where firms broadcast to a very small number of listeners.